



ELI5 · CLOUD PHYSICS

Why are clouds white, but rain clouds dark?

REFERENCES	TOKENS	VERIFICATION	COMPILED
13 verified	≈1.7k	Crossref	26 Aug 2026

SUMMARY

A cloud looks white because its tiny clear drops or bits of ice send the Sun's colors back to us mixed together. A thick rain cloud blocks and loses more light before it reaches the bottom, so the part we see can look grey or dark.

01 A cloud is a crowd of clear drops, not a lump of white stuff

- Aircraft measurements show that an ordinary warm-cloud drop is about **22 millionths of a metre across**—much wider than a wave of visible light. [Fu et al. 2022](#), [Kim et al. 2003](#)
- A drop this big sends red, green, and blue light away with almost the same strength, although their exact directions differ. [Kokhanovsky 2004](#), [Bréon & Goloub 1998](#)

- High, icy clouds work similarly. Their ice pieces send light in many directions, with shape and roughness changing the pattern. [Yang et al. 2013](#), [Yang et al. 2018](#)

02 Many changes of direction make a soft white glow

- A cloud's crowd of clear drops turns sunlight again and again. Those trips mix the Sun's colors and spread a soft white glow instead of one tiny reflection. [Bohren 1987](#), [Davis & Marshak 2010](#), [Kokhanovsky 2004](#)
- You can see the light paths in [Figure 1](#).

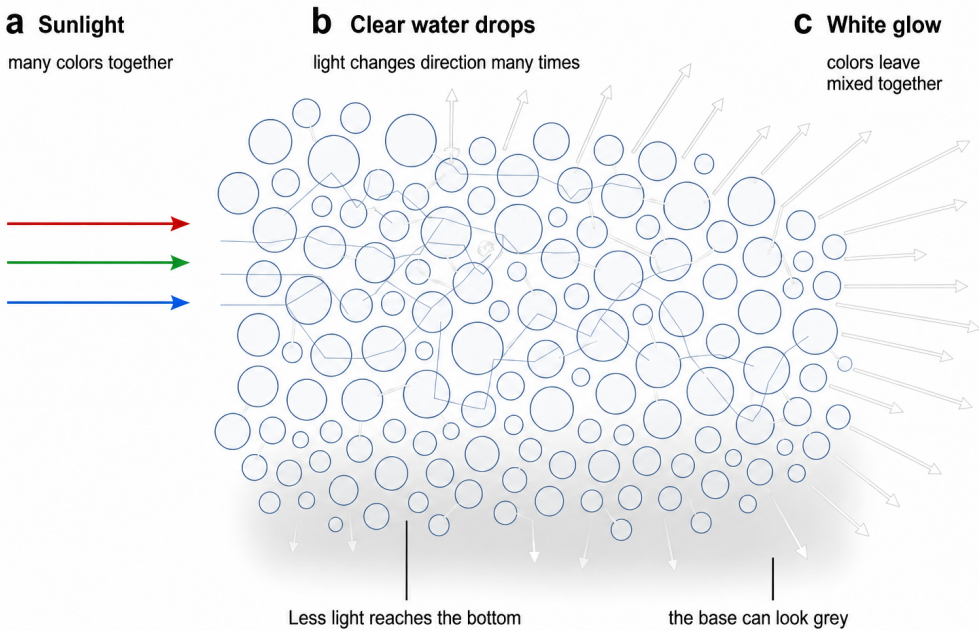


Figure 1. A crowd of clear drops makes a white cloud.

- Sunlight goes into the cloud. The drops send it in many new directions.
- Much of the light escapes as a white glow. Less reaches the bottom of a thick cloud.
- The lines show the main idea. Real light does not follow these exact drawn paths.
- Sources: [Bohren 1987](#), [Kokhanovsky 2004](#), [Kim et al. 2003](#), [Kato & Marshak 2009](#)

**03 A thick cloud can shade its own bottom**

- A cloud gets harder for light to cross when it holds more water, has more drops, or gives light a longer path. On **13 overcast days**, the amount of liquid water explained most of the changes in how hard the cloud was for light to cross. [Kim et al. 2003](#), [Brenguier et al. 2000](#)
- In a deep cloud, lots of light escapes from the top and sides before it reaches the bottom. The base then looks grey because it is dimmer, not because the drops turned grey or black. [Bohren 1987](#), [Kokhanovsky 2004](#)
- Real clouds are lumpy. The Sun's place and your viewing place also change which bright and shaded parts you see. [Kato & Marshak 2009](#), [Fu et al. 2022](#)

04 White is the usual answer, but clouds can show faint colors

- Thick overcast was not a perfectly colorless filter in measured daylight. Many trips through water drops soaked up a tiny bit more red light, so the light below became a little bluer. [Lee & Hernández-Andrés 2005](#)
- Thin clouds with drops of nearly one size can sort colors into rings near the Sun. As the cloud gets thicker, those colors wash into a bright grey background. [Gedzelman & Lock 2003](#)
- Some cloud edges show narrow rainbow-like bands. A recent study found that the bands can come from drop size changing along the edge. [Laven & Hall 2026](#)

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